SQLite primer: Question sheet

Exercise 1

Write an SQL query that returns the `patient_id`, `first_name`, `surname` and `date_of_admission` for all patients admitted on or after 1st January 2026. Order the results by admission date, with the most recently admitted patients first.

Exercise 2

Write a query to identify patients who have both:

- A systolic blood pressure (`sys_bp`) of 140 mmHg or higher.
- A heart rate (`heart_rate`) greater than 100 beats per minute.

Return `patient_id`, `sys_bp`, `dia_bp` and `heart_rate`. Order the results by systolic blood pressure, highest first.

Exercise 3

Using the `medications` table, write a query to determine how many times each drug has been prescribed.

Your query should:

- Return the drug name and number of prescriptions.
- Use the alias `prescription_count` for the calculated column.
- Order the results from the most frequently prescribed drug to the least

Exercise 4

The `patient_data` table contains demographic information, while the `medications` table contains prescribed drugs. Write a query that returns the following information for patients prescribed Amlodipine:

- Patient ID
- First name and surname
- Date of birth
- Prescribed drug name
- Dose, units and frequency

Order the results alphabetically by surname.

(c) Dr Alan Davies – Health Data Works Ltd.

Exercise 5

A researcher wants to investigate the medications prescribed to patients with elevated systolic blood pressure. Write an SQL query that combines information from all three tables (`patient_data`, `admission_observations` and `medications`).

Your query should:

1. Return the patient's ID, first name and surname.
2. Include their systolic and diastolic blood pressure.
3. Include their prescribed medication name, dose and frequency.
4. Include only patients with a systolic blood pressure of 160 mmHg or higher.
5. Add a computed column called `bp_category` using `CASE`, classifying systolic BP as:
    - 160–179: Grade 2 Hypertension
    - 180 or higher: Grade 3 / Severe Hypertension
6. Order the results by systolic blood pressure, highest first.

Exercise 6

Using the database schema (click database structure) and the https://dbdiagram.io/d tool (which uses DBML (Database Markup Language)), create a data model (entity relationship model) of the database showing the three tables and the relationships between them.

Exercise 7

Write an SQL query that returns all medications whose prescribed drug name contains the text morph, regardless of where it appears in the name. Return the `medication_id`, `patient_id`, `prescribed_drug_name`, `strength` and `dose`.

Exercise 8

Write an SQL query that counts how many patients were admitted in each calendar year. Your query should return the admission year and the number of patients admitted in that year, ordered from the most recent year to the earliest.

Exercise 9

Write an SQL query that calculates the average prescribed dose for each medication. Return the drug name, dose units and average dose, rounded to two decimal places. Only include medications that have been prescribed at least three times. Order the results by average dose, highest first.

(c) Dr Alan Davies – Health Data Works Ltd.

Exercise 10

A researcher wants to investigate prescribing patterns for drugs containing the text ‘pril’ (for example, ramipril or lisinopril).

Write an SQL query that:

1. Identifies medications whose names contain ‘pril’.
2. Joins the `medications` and `admission_observations` tables using `patient_id`.
3. Groups the results by prescribed drug name.
4. Returns the drug name, number of prescriptions and average systolic blood pressure.
5. Rounds the average systolic blood pressure to one decimal place.
6. Orders the results by prescription count, highest first.

(c) Dr Alan Davies – Health Data Works Ltd.